

PREREQUISITE REVIEW: LINEAR ALGEBRA

Neural Networks

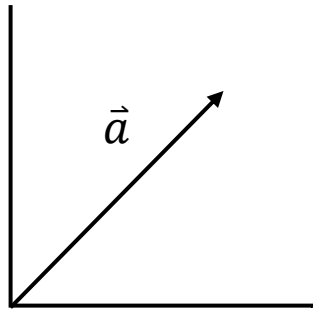
LINEAR ALGEBRA

- Concise description of multi-dimensional objects
- Replace high-dimensional geometry problems with algebra problems

VECTORS

Definition:

Quantities with direction and magnitude



$$\vec{a} = \begin{bmatrix} x \\ y \end{bmatrix}$$

BASIC VECTOR OPERATIONS

Sum of vectors:

$$\vec{a} + \vec{b} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} x + u \\ y + v \\ z + w \end{bmatrix}$$

Scalar multiplication:

$$c\vec{a} = c \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} cx \\ cy \\ cz \end{bmatrix}$$

Inner product:

$$\vec{a} \cdot \vec{b} = \begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = xu + yv + wz$$

VECTOR NORMS

Measure the size of a vector

$$\|\vec{a}\|_p = \left(\sum_i |a_i|^p \right)^{\frac{1}{p}}$$

Most commonly

$$\|\vec{a}\|_1 = \sum_i |a_i|$$
$$\|\vec{a}\|_2 = \sqrt{\sum_i |a_i|^2}$$

SPECIAL VECTORS

Unit Vector:

A vector $\vec{a}$ whose norm $\|\vec{a}\| = 1$.

Orthogonal:

Two vectors $\vec{a}, \vec{b}$ whose inner product $\vec{a} \cdot \vec{b} = 0$.

MATRICES

Definition:

A 2D array of numbers

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

BASIC MATRIX OPERATIONS

Transposition:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow A^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

Addition:

$$A + B = \begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} a + e & b + f \\ c + g & d + h \end{bmatrix}$$

Scalar multiplication:

$$eA = e \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} ae & be \\ ce & de \end{bmatrix}$$

Multiplication:

$$AB = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{bmatrix}$$

MATRIX NORMS

Frobenius norm is most common

$$\|A\|_2 = \sqrt{\sum_{i,j} |A_{i,j}|^2}$$

SPECIAL MATRICES

Diagonal matrix:

A square matrix D for which $D_{i,j} = 0$ if $i \neq j$

Identity matrix:

A diagonal matrix I where all non-zero elements are 1.

Symmetric matrix:

A matrix A for which $A = A^T$

Orthogonal matrix:

A matrix A for which $AA^T = I$

TENSORS

Definition:

An ND array of numbers (e.g. 3D)

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}; \begin{bmatrix} e & f \\ g & h \end{bmatrix}$$

BASIC TENSOR OPERATIONS

Addition:

$$A + B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}; \begin{bmatrix} e & f \\ g & h \end{bmatrix} + \begin{bmatrix} i & j \\ k & l \end{bmatrix}; \begin{bmatrix} m & n \\ o & p \end{bmatrix} = \begin{bmatrix} a + i & b + j \\ c + k & d + l \end{bmatrix}; \begin{bmatrix} e + m & f + n \\ g + o & h + p \end{bmatrix}$$

Scalar multiplication:

$$iA = i \begin{bmatrix} a & b \\ c & d \end{bmatrix}; \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ai & bi \\ ci & di \end{bmatrix}; \begin{bmatrix} ei & fi \\ gi & hi \end{bmatrix}$$

Multiplication:

There are several different ways, but get messy...

MATRIX-VECTOR MULTIPLICATION

- Treat the vector like matrix with one column

$$A\vec{v} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix}$$

TRANSFORMATIONS

Scaling:

$$A\vec{v} = \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax \\ dy \end{bmatrix}$$

Rotation:

$$A\vec{v} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \end{bmatrix}$$

Linear:

$$A\vec{v} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix}$$

LINEAR SYSTEMS

Assuming A and $\vec{b}$ are known...

$$A\vec{x} = \vec{b}$$

Or more explicitly...

$$\begin{bmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$A_{1,1}x_1 + A_{1,2}x_2 = b_1$$

$$A_{2,1}x_1 + A_{2,2}x_2 = b_2$$

INVERSION FOR LINEAR SYSTEMS



For a matrix A , its inverse is called A^{-1} , and is a matrix satisfying

$$AA^{-1} = I$$

We can solve a linear system $A\vec{x} = \vec{b}$ as

$$\vec{x} = A^{-1}\vec{b}$$

LINEAR DEPENDENCE

Linear combination:

A linear combination of a set of vectors $\vec{a}_1, \vec{a}_2, \vec{a}_3, \dots$ is a sum of scalar multiples of these vectors $c_1\vec{a}_1 + c_2\vec{a}_2 + c_3\vec{a}_3 + \dots$

Linear independent:

A set of vectors $\vec{a}_1, \vec{a}_2, \vec{a}_3, \dots$ for which no vectors can be written as a linear combination of other vectors.

LINEAR DEPENDENCE IN LINEAR SYSTEMS



Carleton
UNIVERSITY

A linear system $A\vec{x} = \vec{b}$ has a unique solution if the number of linearly independent columns in A is equal to the number of elements in $\vec{b}$.

EIGENDECOMPOSITION

- An efficient way of decomposing a square matrix into its parts
- Useful to understand the properties of matrices

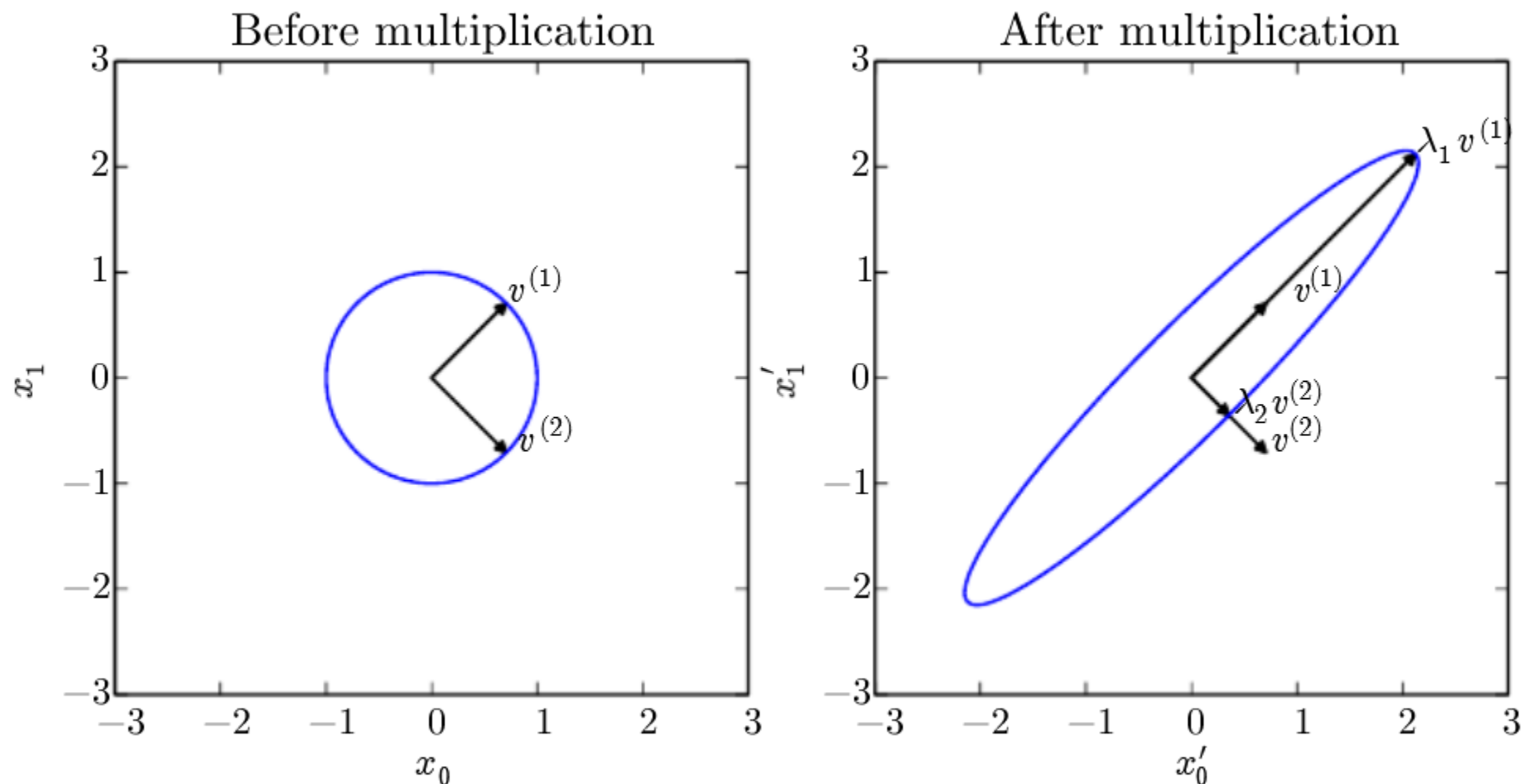
Eigenvectors:

Solutions $\vec{v}$ to the system $A\vec{v} = \lambda\vec{v}$.

Eigenvalues:

Coefficients λ associated with eigenvalues

ILLUSTRATION OF EIGENVALUES



EIGENDECOMPOSITION

If V is a matrix whose columns are eigenvectors of A

If Λ is a diagonal matrix whose elements are eigenvalues of A

The eigendecomposition a matrix A is given by

$$A = V\Lambda V^{-1}$$

If A is symmetric, then V is orthogonal

$$A = V\Lambda V^T$$

TRACE AND DETERMINANT

- Quantify how big a matrix

Trace:

Sum of diagonal elements (or sum of eigenvalues)

$$\text{Tr}(A) = \sum_i A_{i,i} = \sum_i \lambda_i$$

Determinant:

Product of eigenvalues

$$\det(A) = \prod_i \lambda_i$$

SINGULAR VALUE DECOMPOSITION

- Similar to eigendecomposition, but works for non-square matrices

Right singular vectors:

Eigenvectors of $A^T A$

Left singular vectors:

Eigenvectors of AA^T

Singular values:

Square roots of eigenvalues of $A^T A$ or AA^T

SINGULAR VALUE DECOMPOSITION



If V is a matrix whose columns are right singular vectors of A

If U is a matrix whose columns are left singular vectors of A

If D is a diagonal matrix whose elements are singular values of A

The singular value decomposition a matrix A is given by

$$A = UDV^T$$

MATRIX PSEUDOINVERSE

Assuming A and $\vec{b}$ are known...

$$A\vec{x} = \vec{b}$$

Want to find some matrix A^+ such that

$$\vec{x} = A^+\vec{b}$$

MATRIX PSEUDOINVERSE

If V is a matrix whose columns are right singular vectors of A

If U is a matrix whose columns are left singular vectors of A

If D^+ is a diagonal matrix whose elements are the reciprocals of the singular values of A

The Moore-Penrose pseudoinverse of a matrix A is given by:

$$A^+ = VD^+U^T$$

MATRIX PSEUDOINVERSE

The solution to our system is:

$$\vec{x} = A^+ \vec{b}$$

If the system is underdetermined, $\vec{x}$ is the solution with smallest L2 norm.

If the system is overdetermined, $\vec{x}$ gives us the closest possible solution.